

WEEK 4 – ASSIGNMENT

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group

Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file

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Task 3: ADF Basics

Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity

Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design

Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)

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Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment

Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully
- Data copied to destination
- Metadata validated